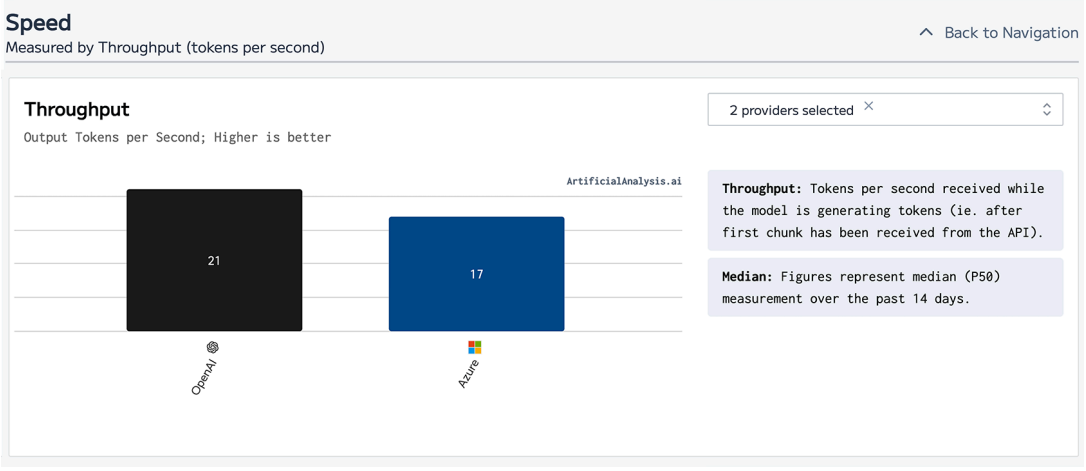


Figure 3.2 Speed comparison of GPT-4 models served from OpenAI versus Azure. The output token throughput is within a mere couple of dozen tokens per second maximum. (Source https://artificialanalysis.ai)

3.2.1 Challenges and how the SDK solves streaming in web applications

While the benefits of streaming are clear, implementing it from scratch presents several technical challenges. The Vercel AI SDK is designed to abstract away these complexities, allowing developers to enable streaming with minimal effort:

- *Asynchronous processing*—Streaming requires handling a continuous flow of data chunks. The SDK's API is built on asynchronous operations, providing a familiar and streamlined way to handle this data flow.
- *Connection management*—The SDK manages the live connection to the server on your behalf, gracefully handling interruptions and reconnections to ensure an uninterrupted experience.
- *Data buffering*—The SDK's built-in hooks and utilities efficiently manage the data as it arrives, taking care of buffering and processing so the UI can be updated smoothly.
- *Error handling*—The SDK provides a centralized and consistent way to handle errors that might occur midstream, allowing you to provide clear feedback to the user without breaking the experience.

How does streaming work, exactly? I will explain this process with some technical details and some diagrams to make it easier to grasp.

HOW STREAMING WORKS

Streaming is a way of transmitting data incrementally rather than all at once. For AI-generated responses, this means content is delivered to the user in real time as the AI model produces it.